

SUMMARY OF QUALIFICATIONS

- Ph.D. in Construction Engineering at École de technologie supérieure
- Experience structuring ambiguous technical problems into methodologies, quantitative analyses, visual frameworks, and evidence-based conclusions
- Strong background in research synthesis, experimental design, data interpretation, model-based reasoning, sustainability-oriented innovation, and technical communication
- Developed peer-reviewed manuscripts, conference presentations, visual materials, and analytical frameworks for specialized and broader audiences
- Ability to work in a team and under pressure
- Adherence to deadlines and ability to adapt and work independently
- International research experience in Canada, Brazil, and Australia
- Fluent in French, English, and Portuguese

CORE SKILLS

Analytical and consulting-relevant skills: Problem structuring, research synthesis, methodology development, quantitative data analysis, experimental design, performance evaluation, trade-off analysis, data visualization, technical writing, presentation development

Technical and digital skills: Sustainability-oriented construction materials, optical characterization, mechanical testing, image processing, optoelectronic systems, smart materials, GIS-based environmental risk analysis

Tools: Microsoft PowerPoint, Microsoft Excel, Microsoft Word, Python, QGIS, JOSM, LaTeX, Canva, Inkscape

EDUCATION

Ph.D. in Construction Engineering: École de technologie supérieure | Montréal, QC | 2022 to 2026 | Thesis: Design and performance of fiber-optic light-transmitting cementitious composites | Expected completion: July 2026

M.Sc. in Civil Engineering: Sao Paulo State University | Brazil | 2020 to 2022 | Thesis: Influence of aggregate sizes and packing combination on the properties of pervious concrete

B.Sc. in Civil Engineering: State University of Mato Grosso | Brazil | 2014 to 2019

RESEARCH AND PROFESSIONAL EXPERIENCE

Ph.D. Researcher, Construction Engineering | École de technologie supérieure | Montréal, QC | 2022 to 2026

- Led a multi-year doctoral project on optical fiber-embedded cementitious composites, combining low-carbon material design, optical characterization, mechanical performance evaluation, privacy-preserving light transmission, and interactive material applications
- Structured an emerging research problem into measurable objectives, experimental protocols, analytical models, and visual frameworks
- Designed and executed protocols to evaluate how material configurations and testing conditions influence optical and mechanical performance
- Analyzed datasets involving optical transmittance, fiber distribution, mechanical response, digital image reconstruction, and interactive sensing behavior

- Developed models and data visualizations to compare configurations, interpret performance trade-offs, and communicate results clearly
- Synthesized fragmented scientific literature to identify methodological gaps and propose clearer classification and evaluation frameworks
- Prepared peer-reviewed manuscripts, conference papers, technical presentations, and visual materials for specialized and broader audiences
- Collaborated with supervisors and researchers across construction materials, optics, sensing, sustainability, and smart systems

Research Intern | University of Sydney | Sydney, Australia | *Dec. 2021 to Feb. 2022*

- Contributed to a flood risk assessment project in Indonesia using GIS-based spatial analysis with QGIS and JOSM
- Interpreted geospatial information related to infrastructure exposure, environmental risk, climate-related flooding, and sea-level rise

Teaching Assistant, Building Construction | Sao Paulo State University | Brazil | *Mar. 2021 to Sept. 2021*

- Supported a 60-hour undergraduate course and explained applied engineering concepts to students with different levels of preparation

Civil Engineering Intern | Municipality of Tangara da Serra | Brazil | *2018*

- Supported technical activities within the municipal infrastructure department, including engineering documentation and public-sector construction activities

SELECTED RESEARCH OUTPUTS

- First-author publications in Measurement Science and Technology and Scientific Reports on optical characterization, light-transmitting cementitious composites, privacy-preserving light transmission, and digital image reconstruction
- Conference presentations at Canadian Society of Civil Engineering, International Society for Computing in Civil and Building Engineering, and Centre for Optics, Photonics, and Lasers
- Co-author of peer-reviewed work in Journal of Building Engineering and Sustainability on sustainable pervious concrete and material performance

AWARDS, PRESENTATIONS AND VISUAL COMMUNICATION

- PBEEE Merit Scholarship for Foreign Students, FRQNT, 2025 to 2026
- Doctoral Scholarship, École de technologie supérieure, 2022 to 2026
- Summer Research Scholarship, University of Sydney, 2022
- NSERC Science Exposed, Jury Prize for scientific photography, 2024
- Acfas la preuve par l'image, selected among 20 images exhibited at the Montreal Biodome, 2026
- ÉTS Photo Contest, Arts and Culture category, 2025
- Art-science exhibition participant, Collectif INTRIGO, Amos, QC, 2025, Presentation of light-transmitting cementitious composites as immersive installations in an exhibition exploring the translation of quantum sciences into sensory experiences